

⌈ Rationalize.

Let  $K$  be a field. Given  $\theta \in K$ , find an irreducible polynomial

$$P(x) = p_0 x^n + p_{n-1} x^{n-1} + \dots + p_1 x + p_0 \quad \text{s.t.} \quad P(\theta) = 0.$$

Then,  $\theta^{-1} = -p_0^{-1} \cdot (p_n \theta^{n-1} + p_{n-1} \theta^{n-2} + \dots + p_1) \in K$  is the multiplicative inverse of  $\theta$ .

⌈ Problem. Rationalize the denominator of  $\frac{1}{3 + 2 \cdot \sqrt[3]{2} + \sqrt[3]{4}}$ .

Proof. In other words, find  $(3 + 2 \cdot \sqrt[3]{2} + \sqrt[3]{4})^{-1}$  as  $\mathbb{Q}$ -linear combination for  $\sqrt[3]{2}$ .

Let's calculate, by the Euclidean Algorithm,

$$x^3 - 2 = (x^2 + 2x + 3) \cdot (x - 2) + (x + 4)$$

$$x^2 + 2x + 3 = (x + 4)(x - 2) + 11$$

$$x + 4 = 11 \cdot \left(\frac{1}{11}x + \frac{4}{11}\right) \quad (\text{Not necessary term})$$

$$\text{So, } (x + 4) = (x^3 - 2) - (x^2 + 2x + 3) \cdot (x - 2)$$

⌋ Put

$$(x + 4)(x - 2) + 11 = x^2 + 2x + 3$$

$$\text{Then, we obtain } (x - 2) \cdot (x^3 - 2) - (x^2 + 2x + 3) \cdot (x^2 - 4x + 5) = -11.$$

Since  $\theta = \sqrt[3]{2}$  satisfies that  $\theta^3 - 2 = 0$ , so

$$(\theta^2 + 2\theta + 3) \cdot (\theta^2 - 4\theta + 5) = 11, \text{ or}$$

$$(3 + 2 \cdot \sqrt[3]{2} + \sqrt[3]{4})^{-1} = \frac{1}{11} \cdot (\sqrt[3]{4} - 4 \cdot \sqrt[3]{2} + 5). \quad \square$$

⌈ In general, let  $P(x)$  be an irreducible polynomial over  $F$ .

Then,  $K = F[x]/(P(x))$  has a root of  $P(x)$ , namely  $\alpha$ .

For  $F$ -polynomial  $f(x)$ , the inverse of  $f(\alpha) \neq 0$  is:

$$(f(x) + (P(x)) = 1), \text{ since } F[x] \text{ is P.I.D and } P(x) \text{ is irreducible,}$$

So,  $a(x)f(x) + b(x)P(x) = 1$  for some  $a(x), b(x) \in F[x]$ ,

$$a(\alpha) \cdot f(\alpha) + b(\alpha) \cdot P(\alpha) = a(\alpha) \cdot f(\alpha) + 0 = 1, \text{ or}$$

$$f(\alpha)^{-1} = a(\alpha).$$